

Addis Ababa University
Addis Ababa Institute of Technology
School of Mechanical and Industrial Engineering

Engineering Materials I
Course Outline

Course: Engineering Materials I
Course Code: MEng 2122
Course Owner: Manufacturing Chair
Course Duration: One Semester
Credit Hours: 3
ECTS: 5

Contact Hours: Two Lec. & Tuto. each & lab
Grade Level: 2nd Year
Acad. Year: 2014 E.C (2nd Semester)
Course Instructor: Ayalsew Dagne
Office: 314 C
Contact address: ayoudag@gmail.com

Course Objectives:

- Understand the main concepts of engineering materials and their advancements.
- Identify & understand the types of materials and their specific applications.
- Identify & understand the influence of structural crystallinity on the properties of materials.
- Understand the type of defects and identify their influence on the properties of crystal structures.
- Understand laws of diffusion and explain application of diffusion in engineering materials.
- Identify causes and main types of plastic deformation, mechanical properties and types of metal testing.
- Identify & understand failure and mechanism of fracture.
- Understand the main concepts of phase diagram and phase transformations for different alloys.

Course Contents:

Chapter – 1: Introduction

- 1.1. Historical Background
- 1.2. Importance of Studying Engineering Materials
- 1.3. Classification of Materials
- 1.4. Advancement of Modern Materials
- 1.5. Overall Correlations

Chapter – 2: Atomic Structure & Interatomic Bonding

- 2.1. Introduction
- 2.2. Atomic Structure
- 2.3. Atomic Bonding in Solids

Chapter – 3: Structure of Crystalline Solids

- 3.1. Introduction
- 3.2. Crystal Structures
- 3.3. Crystallographic Points, Directions & Planes
- 3.4. Crystalline & Non-Crystalline Solid Materials

Chapter – 4: Imperfections in Solids

- 4.1. Introduction
- 4.2. Types of Defects
- 4.3. Miscellaneous Imperfections
- 4.4. Microscopic Examination

Chapter – 5: Diffusion

- 5.1. Introduction
- 5.2. Diffusion Mechanisms
- 5.3. Fick's Law

- 5.4. Factors that Affect Diffusion
- 5.5. Diffusion in Semiconducting Materials
- 5.6. Other Diffusion Paths

Chapter – 6: Mechanical Properties of Metals

- 6.1. Introduction
- 6.2. Elastic Deformation
- 6.3. Plastic Deformation
- 6.4. Property Variability and Design/Safety Factors

Chapter – 7: Dislocation and Strengthening Mechanisms

- 7.1. Introduction
- 7.2. Dislocations & Plastic Deformation
- 7.3. Mechanisms of Strengthening in Metals
- 7.4. Recovery, Recrystallization & Grain-Growth

Chapter – 8: Failure and Mechanisms of Fracture

- 8.1. Introduction
- 8.2. Fracture
- 8.3. Fatigue
- 8.4. Creep

Chapter – 9: Phase Diagram

- 9.1. Introduction to Phase Diagrams
- 9.2. Basic Concepts
- 9.3. Binary Phase Diagrams
- 9.4. Iron-Carbon System

Laboratory Practices: Tensile, hardness, impact & torsion tests of metals.

Assessment: Assignment: 10%, Quiz: 10%, Lab report: 10%, Intermediate exam (2): 30%, & Final exam: 40%

Attendance: Minimum of 75% attendance during both tutorial and lecture hours is mandatory. 100% presence during laboratory activities and exams. Absence of classes without any recognized reason has no acceptance.

Reading Materials

Text Book:

1. William D. Callister, *“Material Science and Engineering: An Introduction”*, 9th edition, John Wiley & Sons, 2013.

Reference:

1. Brian S. Mitchell, *“An Introduction to Materials Engineering and Science for Chemical and Materials Engineers”*, 2004, Wiley Interscience.
2. Veron John, *“Introduction to Engineering Materials,”* Macmillan Education UK, May 9, 2003.
3. Michael F. Ashby, Hugh Shercliff, David Cebon, *Materials: Engineering, Science, Processing and Design*, 3rd edition, Elsevier Ltd, 2014.
4. William D. Callister, *“Fundamentals of Material Science and Engineering” – An Integrated Approach*, 3rd edition, John Wiley & Sons, 2008.